Condition	β	Budget	Loss	Pref. Acc.	KL/Token	RM Score	Length	At Cap
Standard	0.10	1 epoch (91 steps)	0.674	0.628	1.40×10^{-4}	1.38 ± 1.16	155.2 ± 104.6	43.1%
Short fork	0.03	581 pairs (37 steps)	0.691	0.607	-1.02×10^{-4}	1.31 ± 1.26	153.7 ± 105.7	43.8%
Short fork	0.10	581 pairs (37 steps)	0.686	0.624	-2.04×10^{-4}	1.29 ± 1.21	156.7 ± 104.2	43.8%
Short fork	0.30	581 pairs (37 steps)	0.677	0.624	-4.50×10^{-5}	1.31 ± 1.17	151.7 ± 105.2	42.4%

Table 1: DPO β ablation. The standard model was trained for one epoch. the short forks used the same 581 retained training pairs and optimizer budget. KL/token is the sampled-response estimator averaged over response tokens. Reward-model score and generated length are mean $\pm$ standard deviation over 290 held-out prompts.

Methods of Pos Training Alignment in LLMs

Anonymous Author(s)
Affiliation
Address
email

Abstract

In this paper we look at various forms of post training for LLMs and study their effect on our Models. We will be doing 3 Main experiments: Direct Preference Optimization (DPO), Proximal Policy Optimization (PPO), Group Relative Policy Optimization (GRPO) and then we compare their performances on Safe and Unsafe prompts. We will also see the differences between Reinforcement Learning with Verifiable Rewards (RLVR) and Reinforcement Learning from AI Feedback (RLAIF).

1 Introduction

This is a short study done on the different methods of LLM alignment with 3 main focuses. DPO, PPO, and GRPO. For each method we will use collect and present our data with a short analysis of our results. Alongside that we will also explore how much of a differences using RLVR vs RLAIF makes.

2 Methods and Setup

2.1 Direct Preference Optimization (DPO)

- **Dataset:** UltraFeedback Preference Pairs (course provided)
- **Model:** Qwen2.5-1.5B-Instruct (Forzen with LoRA)

3 Results

3.1 Direct Preference Optimization (DPO)

We first do a standard full training run for DPO, then we make three short forks for a β Study. The data can bee seen in Table1. We can see a very similar level of preference accuracy, with the exception

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Model	Preferred Longer	Length Matched	Rejected Longer	Overall	DPO Loss
Standard	44.9% (78)	57.5% (80)	67.1% (79)	56.5% (237)	0.6821
Length-balanced	43.6% (78)	58.8% (80)	78.5% (79)	60.3% (237)	0.6760

Table 2: Held-out DPO preference accuracy stratified by the relative lengths of preferred and rejected responses. Counts are shown in parentheses. Nine of the 246 held-out pairs were excluded by the prompt-length filter.

Quantity	Value
Wall-clock time	293.8 s
Peak VRAM (allocated)	4.91 GB
Mean reward, first 5 / last 5 updates	1.14 / 0.53
Mean reward, all 20 updates	1.24 ± 1.58
Responses at 512-token cap without EOS	4 / 20
Maximum clip fraction	0.000
Critic explained variance (mean)	-15.2
Critic–return correlation (mean)	0.089
Held-out reward: midpoint / PPO	1.866 / 1.838
Held-out reward difference (SEM)	-0.028 (0.035)
Held-out response length: midpoint / PPO	287.5 / 289.0

Table 3: Standard PPO continuation and held-out evaluation on 100 common prompts. Training statistics cover 20 updates; held-out responses use greedy decoding with a 768-token cap.

of $\beta = 0.03$ which is the lowest at 0.607. This tells us that too low a value will damage performance, but we cant conclusively say a best value with our data. The Response Lengths are also very similar, however considering that all rows show over 40% of responses exceed the length cap (and were hence truncated), this data cant not be meaningfully interpreted.

We then try to determine if the model is being biased with its respond length and learning a false correlation between length and preferred response. We can see in Table2 that overall the model trained on length balanced data performs marginally better than standard. With Length Matched pairs, the accuracy is very close to each other. But for Rejected response being longer, the length balanced model does much better than standard, even though performance in preferred response longer is quite similar. This shows us that improvement is not uniform across strata, and while length balancing does help with preference discrimination in, it does not conclusively demonstrate that the standard model learned a length bias

Policy	ϵ	Held-out reward (SEM)	ΔReward (SEM)	Mean length
Standard PPO	0.20	1.838 (0.125)	-0.028 (0.035)	289.0
Short fork	0.05	1.867 (0.129)	$+0.001$ (0.026)	289.3
Short fork	0.20	1.846 (0.127)	-0.020 (0.033)	288.9
Short fork	0.50	1.847 (0.129)	-0.019 (0.030)	289.2

Table 4: Held-out evaluation of PPO clipping forks on the same 100 prompts. ΔReward is the paired difference against the untouched midpoint; SEM denotes standard error of the mean.

β_{KL}	Train reward	First 4 / last 4	Final KL	Mean length	EOS rate
0.00	0.92	1.64/0.20	0.0054	253.0	62%
0.10	1.51	1.66/1.35	0.0048	243.6	75%
0.20	1.54	1.66/1.41	0.0025	247.9	75%

Table 5: KL-pressure forks over eight updates each. Train reward is the reward-model score on training rollouts; final KL is the reported sampled estimator.

3.2 Proximal Policy Optimization (PPO)

3.3 Group Relative Policy Optimization (GRPO)

3.4 Safety Calibration of the Aligned Policies

3.5 Feedback Source: RLVR versus RLAIF

3.6 Cross Task Discussion

4 Conclusions

5 References

Appendix

A More Method Details

B Deviations and Caveats

B.1 Direct Preference Optimization (DPO)

- We changed the Batch Size to 1 and Gradient Accumulate Steps to 16, keeping effective batch size constant. This was done as our hardware would run out of memory otherwise.